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Wheel Assmbly Force Analysis

Initial Factors

Factor of Safety

Minimum acceptable factor of safety for this subsystem: $FOS := 1.7$

Lifetime Analysis

The car must be designed to last $D := 2000\ \text{km}$

Calculate the lifetime cycles of the wheels and hubs

Tire Diameter: $d_T := 4.134\ \text{in} = 0.105\ \text{m}$

Tire Circumfrence: $c_T := d_T \cdot \pi$

Lifetime Cycles: $L_C := \frac{D}{c_T} = 6.0628 \cdot 10^{6}$

Load Cases

Pure Longitudingal

Max Brake:

The following values were provided by the Brakes subassembly team

Accelration: $(-1.60)\ g_e = -15.6906\ \frac{\text{m}}{\text{s}^2}$

Front

Front Corner Brake Force: $F_{BF} := \frac{100\ \text{N}}{2} = 50\ \text{N}$

Brake Torque at axis: $T_B := F_{BF} \cdot \frac{d_T}{2} = 2.6251\ \text{N m}$

Rear

Rear Brake Force: $F_{BR} := \frac{100\ \text{N}}{2}$

Proportioned Rear Brake Force: $F_{BRP} := \frac{100\ \text{N}}{2}$

These values are applied on the tire at the road, directly below brake rotor, so 18" away from center axis

Max Acceleration

The following values were provided by the drivetrain subassembly team

Acceleration: $a_{accel} := 1.5\ g_e = 14.71\ \frac{\text{m}}{\text{s}^2}$

Max Half Shaft Torque: $T_{HS} := 1.306\ \text{N m}$

Longitudinal Accel Force:

$F_{accel} := \frac{T_{HS}}{4.134\ \text{in}} = 12.44\ \text{N}$

Our goal is to achieve a 4 second 75m accel time

Check to see if accel time is a liberal enough guess:

$d := 75\ \text{m}$

$\frac{1}{2} \cdot a_{accel} \cdot t^2 = d$

$t := \sqrt{\frac{d}{\frac{a_{accel}}{2}}} = 3.1933\ \text{s}$

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Pure Lateral

Max Lateral Slide

The maximum lateral slide acceleration is assumed to be the same as the maximum longitudinal breaking acceleration

Acceleration assumed equal at all four tires

Car mass: $m_C := 3\ \mathrm{kg}$

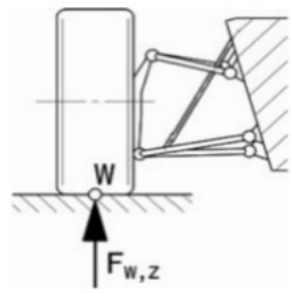
Max Lateral Accel: $a_{Lat} := (-1.60)\ g_e = -15.69\ \frac{\mathrm{m}}{\mathrm{s}^2}$

Lateral Force: $F_{LS} := a_{Lat} \cdot m_C = -47.07\ \mathrm{N}$

Force applied latterally at tire contact with ground

Pure Vertical

Static



Car mass: $m_C := 3\ \mathrm{kg}$

Weight proportion front: $Ant_F := 0.40$

Weight proportion rear: $Ant_R := 0.60$

Static Front Wheel Load: $F_{wzf0} := \frac{m_C}{2} \cdot Ant_F\ g_e = 5.884\ \mathrm{N}$

Static Rear Wheel Load: $F_{wzr0} := \frac{m_C}{2} \cdot Ant_R\ g_e = 8.826\ \mathrm{N}$

Dynamic

Max Bump Accelertion: $a_{Bump} := 5\ g_e = 49.03\ \frac{\mathrm{m}}{\mathrm{s}^2}$

Max Bump Front Wheel Load: $F_{wzf} := \frac{m_C}{2} \cdot Ant_F \cdot a_{Bump} = 29.42\ \mathrm{N}$

Max Bump Rear Wheel Load: $F_{wzr} := \frac{m_C}{2} \cdot Ant_R \cdot a_{Bump} = 44.13\ \mathrm{N}$

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Combined Load Cases

Combined Case 1

Max Break + Positive Max Lat Slide + Max Bump

$F_{x1} := -F_{BF} = -50$ N

$F_{y1} := F_{LS} = -47.07$ N

$F_{z1} := F_{wzf} = 29.42$ N

Combined Case 2

Max Break + Negative Max Lat Slide + Max Bump

$F_{x2} := -F_{BF} = -50$ N

$F_{y2} := -F_{LS} = 47.07$ N

$F_{z2} := F_{wzf} = 29.42$ N

Combined Case 3

Max Accel + Positive Max Lat Slide + Max Bump

$F_{x3} := F_{accel} = 12.44$ N

$F_{y3} := F_{LS} = -47.07$ N

$F_{z3} := F_{wzf} = 29.42$ N

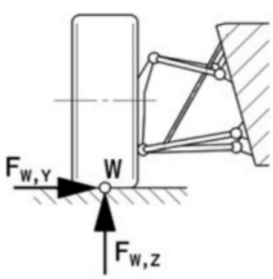
Combined Case 4

Max Accel + Positive Max Lat Slide + Max Bump

$F_{x4} := F_{accel} = 12.44$ N

$F_{y4} := -F_{LS} = 47.07$ N

$F_{z4} := F_{wzf} = 29.42$ N



Front Left

OUTPUT Matrix X - Calculated	
Tie Rod Force (N)	355.91113
LCA (F) Force (N)	-6797.49370
LCA (R) Force (N)	3135.62436
UCA (F) Force (N)	5556.91985
UCA (R) Force (N)	951.45745
Push Rod Force (N)	3934.30380

OUTPUT Matrix X - Calculated	
Tie Rod Force (N)	-1118.41433
LCA (F) Force (N)	-1842.30031
LCA (R) Force (N)	7612.84496
UCA (F) Force (N)	-7664.84057
UCA (R) Force (N)	-8496.06490
Push Rod Force (N)	7249.97039

OUTPUT Matrix X - Calculated	
Tie Rod Force (N)	1045.01049
LCA (F) Force (N)	2106.91801
LCA (R) Force (N)	-6574.68024
UCA (F) Force (N)	1352.92086
UCA (R) Force (N)	4305.03646
Push Rod Force (N)	3027.76495

OUTPUT Matrix X - Calculated	
Tie Rod Force (N)	-429.31497
LCA (F) Force (N)	7062.11141
LCA (R) Force (N)	-2097.45964
UCA (F) Force (N)	-11868.83956
UCA (R) Force (N)	-3239.57099
Push Rod Force (N)	6343.43154

Rear Left

OUTPUT Matrix X - Calculated	
Tie Rod Force (N)	-466.29480
LCA (F) Force (N)	-6175.47197
LCA (R) Force (N)	2786.60950
UCA (F) Force (N)	2212.74038
UCA (R) Force (N)	4894.93051
Push Rod Force (N)	1510.00122

OUTPUT Matrix X - Calculated	
Tie Rod Force (N)	139.85762
LCA (F) Force (N)	-6078.36266
LCA (R) Force (N)	10132.29501
UCA (F) Force (N)	-1716.87416
UCA (R) Force (N)	-15386.01377
Push Rod Force (N)	8997.56113

OUTPUT Matrix X - Calculated	
Tie Rod Force (N)	-88.90945
LCA (F) Force (N)	5882.70938
LCA (R) Force (N)	-7901.23044
UCA (F) Force (N)	-3069.53396
UCA (R) Force (N)	9218.43886
Push Rod Force (N)	1509.92863

OUTPUT Matrix X - Calculated	
Tie Rod Force (N)	517.24296
LCA (F) Force (N)	5979.81868
LCA (R) Force (N)	-555.54493
UCA (F) Force (N)	-6999.14851
UCA (R) Force (N)	-11062.50542
Push Rod Force (N)	8997.48854

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Drive Lug Analysis

Lug Diameter: $d_L := \frac{3}{8}\ \text{in} = 9.525\ \text{mm}$

Lug Mounting Diameter: $d_{LM} := 2.25\ \text{in} = 57.15\ \text{mm}$
Number of Lugs: $n_L := 4$
Tire Diameter: $d_T = 0.105\ \text{m}$
Lug/Hole Contact Depth: $t := 10\ \text{mm}$
Breaking force at road: $F_{long_max} := F_{BF} = 50\ \text{N}$ (Highest worst case longitudinal force)
Brake Torque at axis: $T_B := F_{long_max} \cdot \frac{d_T}{2} = 2.6251\ \text{N m}$

Applied to only one drive lug

Shear Force at drive lugs: $F_{1Lug} := \frac{T_B}{\frac{d_{LM}}{2}} = 91.8667\ \text{N}$

Shear Stress in one lug: $\tau_{1Lug} := \frac{4 \cdot F_{1Lug}}{\pi \cdot \frac{d_L^2}{2}} = 1.2893\ \text{MPa}$

Bearing Stress in one lug: $\sigma_{b1Lug} := \frac{F_{1Lug}}{t \cdot d_L} = 0.9645\ \text{MPa}$

Applied to all four drive lugs

Shear Force at drive lugs: $F_{4Lug} := \frac{F_{1Lug}}{4} = 22.9667\ \text{N}$

Shear Stress in one lug: $\tau_{4Lug} := \frac{4 \cdot F_{4Lug}}{\pi \cdot \frac{d_L^2}{2}} = 0.3223\ \text{MPa}$

Bearing Stress in one lug: $\sigma_{b1Lug} := \frac{F_{4Lug}}{t \cdot d_L} = 0.2411\ \text{MPa}$

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